HPU CONTROL BOX P/N: 88002255



The HPU Control Box from Magswitch is a compact, HMI-based controller for hydraulically actuated magnetic tooling. It drives the directional valve coils that switch a magnet between OPEN and CLOSED, and—when configured for the MTC build—the relief valve coil as well.

A 4.3-inch color touchscreen provides three operating styles: momentary JOG for maintenance, latching TURN ON / TURN OFF for normal use, and an automatic CYCLE test with configurable hold and interval times. Two Hall-effect sensors monitor magnet position, enabling closed-loop confirmation with adjustable thresholds.

All field wiring terminates at two plug-and-play M12 connectors. A dedicated E-STOP, sensor-disconnect detection, a position-timeout alarm and a flashing alarm lamp make the unit safe and easy to operate on the production floor.

General Specifications

Item	Specification
Product Name	HPU Control Box
Part Number (P/N)	88002255
Series / Model	MagPilot B43
Firmware	V3.1.1
Enclosure Dimensions (W × H × D)	158 × 90 × 46 mm (6.22 × 3.54 × 1.81 in)
Display	4.3 in color HMI, 480 × 260 pixels
Supply Voltage	24 V DC
Supply Current	3 A max
Digital Outputs	3 valve-coil outputs — Y1 Valve A (OPEN), Y2 Valve B (CLOSE), Y3 Valve C (MTC only)
Analog Inputs	2 × 0–5 V Hall-sensor inputs (AD0 / AD1)
Position Sensors	2 × 270758 Hall-effect sensors, 24 V, 0–5 V, 3-wire
Connectors	2 × M12, 5-pin, A-coded (Connector A: outputs, Connector B: inputs)
Net Weight	425 g (0.94 lb)
Operating Temperature	-10 °C to +50 °C (14 °F to 122 °F)
Enclosure Rating	IP54 (dust-protected, splash-resistant)

KEY FEATURES

- **4.3-inch color touchscreen** with intuitive, menu-driven screens and a consistent dark/industrial interface.
- **Three control styles —** JOG (momentary, for maintenance), latching TURN ON / TURN OFF for normal use, and an automatic CYCLE test.
- **MAC and MTC operating modes —** MAC (default) drives directional valves A and B; MTC adds relief valve C that follows A and B.
- **Closed-loop and open-loop cycling —** closed-loop uses dual Hall-sensor position feedback; open-loop runs on timers only.
- **Dual redundant Hall sensors —** two sensors monitor the same magnet, with either sensor able to confirm position during cycling.
- **Adjustable OPEN / CLOSE thresholds —** set directly in millivolts (default 700 mV open, 2300 mV close).
- **Configurable timing —** A hold, B hold, A-to-B and B-to-A interval times, all set on the HMI in 0.01 s steps.
- **Cycle counting —** current cycle count, programmable target count, and automatic stop when the target is reached.
- **Sensor-disconnect detection —** a reading below 100 mV is treated as a disconnected sensor and will not falsely show OPEN.
- **Position-timeout alarm —** in closed-loop, if the sensor does not confirm within the hold time, the cycle stops and a fault latches.
- **E-STOP —** immediately de-energizes every output (Y1 / Y2 / Y3).
- **Hardware-style interlock —** coils A and B can never be energized at the same time under any mode.
- **Plug-and-play M12 connectors —** no discrete inputs or analog outputs required.

WHAT'S IN THE CONTROL BOX

The unit combines a color HMI, a PLC, and valve-drive electronics inside a 158 × 90 × 46 mm enclosure. All field connections are grouped on two M12 connectors: Connector A powers and switches the valve coils (common-positive / sinking outputs), while Connector B supplies and reads the two Hall-effect position sensors.



Figure 1 — HMI cover screen: CYCLE TEST and CONTROL (manual) entry buttons

HMI OPERATION — MANUAL CONTROL

The CONTROL screen is used for normal operation and maintenance. It shows live sensor voltages, magnet position, valve status, and all manual controls.



Figure 2 — Manual Control screen (MAC mode shown)

Screen item	Function
SENSOR 1 / SENSOR 2	Live voltage from Hall sensors 1 and 2, in millivolts (mV).
MAGNET OPEN / MAGNET CLOSED	Position status lamps (green).
OPEN THRESHOLD (<) / CLOSE THRESHOLD (>)	Editable position thresholds in mV (default 700 / 2300).
VALVE A / B / C	Valve output status lamps; VALVE C is shown only in MTC mode.
TURN ON / TURN OFF	Latching buttons: TURN ON energizes A to OPEN the magnet; TURN OFF energizes B to CLOSE it.
JOG A / JOG B / JOG C	Momentary buttons for maintenance (JOG C — MTC only).
MAC / MTC	Operating-mode switch (MAC is the default).
E-STOP	Emergency button; immediately cuts all outputs (Y1 / Y2 / Y3).
ALARM	Red lamp that flashes (1 s on/off) on a latched sensor fault or E-STOP.
ALARM RESET	Clears the latched sensor and E-STOP alarms.
CYCLE TEST / HOME	CYCLE TEST opens the cycle screen; HOME returns to the cover screen.

Normal operation

1. Select the mode (MAC by default; MTC for the relief valve).

2. Press TURN ON: coil A energizes and the magnet opens. When the sensor voltage drops below the OPEN threshold, MAGNET OPEN lights.

3. Press TURN OFF: coil B energizes and the magnet closes. When the voltage rises above the CLOSE threshold, MAGNET CLOSED lights.

Note: De-energizing coil A returns the valve spool to center, but the magnet remains open. Coil B must be energized to reverse the magnet and close it.

HMI OPERATION — CYCLE TEST

The CYCLE TEST screen repeatedly opens and closes the magnet, counts complete cycles, and (in closed-loop) confirms each position with the Hall sensors.

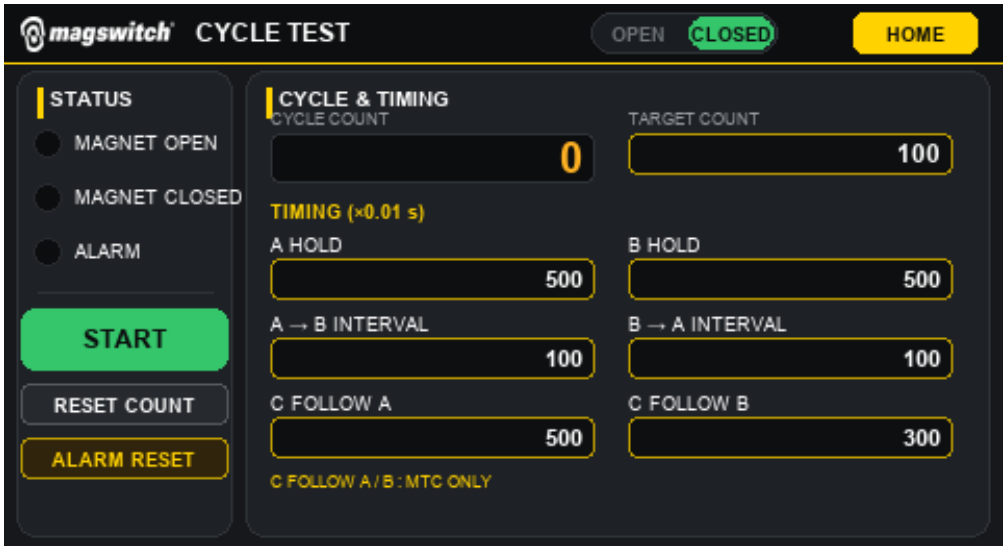


Figure 3 — Cycle Test screen (closed-loop CLOSED selected)

Screen item	Function
OPEN / CLOSED	Loop-mode switch. CLOSED = closed-loop (default, sensor confirmed); OPEN = open-loop (timing only).
STATUS lamps	MAGNET OPEN, MAGNET CLOSED and ALARM indication.
START / STOP	Starts and stops the cycle (green START / red STOP).
RESET COUNT	Stops the cycle and clears the cycle counter.
ALARM RESET	Clears the latched sensor and E-STOP alarms.
CYCLE COUNT	Number of completed cycles (read-only).
TARGET COUNT	The cycle stops automatically when this count is reached.
A HOLD / B HOLD	Energize time for coil A and coil B (×0.01 s).
A → B INTERVAL / B → A INTERVAL	Delay between the A and B strokes (×0.01 s).
C FOLLOW A / C FOLLOW B	Relief-valve C follow time (×0.01 s); MTC only, ignored in MAC mode.

All timing values are entered in 0.01 s steps — for example, 500 = 5.00 s and 100 = 1.00 s.

One complete cycle = coil A actuated once + coil B actuated once.

Closed-loop sequence

1. Coil A energizes for the A HOLD time.
2. If a sensor reaches the OPEN threshold (< 700 mV, and above 100 mV so a disconnected sensor is detected) within the hold time, continue.
3. Coil A de-energizes; the A → B interval runs.
4. Coil B energizes for the B HOLD time; a sensor must reach the CLOSE threshold (> 2300 mV).
5. The B → A interval runs and the cycle count increments by one; the sequence repeats until the target count is reached.

If position is not confirmed within the hold time, the cycle stops and a sensor alarm latches. Open-loop (OPEN) runs by timing only and does not check position.

OPERATING MODES

MAC / MTC modes

Mode	Description
MAC (default)	Drives directional valves A and B only, to OPEN and CLOSE the magnet. Relief valve C is not used.
MTC	Adds relief valve C (Y3), which follows A and B according to the C FOLLOW A / B timers. VALVE C and JOG C appear only in this mode.

Control methods

Method	Behavior
JOG (momentary)	The output is held only while the button is pressed. Used for maintenance and setup.
TURN ON / TURN OFF (latch)	One press opens the magnet (A) and one press closes it (B); the state stays until the next command.
CYCLE (automatic)	Repeats A and B automatically to a target count, in open- or closed-loop.

Closed-loop vs. open-loop

Loop	Behavior
CLOSED (closed-loop, default)	Hall position feedback is required. If the threshold is not reached within the hold time, the cycle stops and a sensor alarm latches.
OPEN (open-loop)	Runs on the configured timers only. No position confirmation and no position sensor alarm (manual / open-loop).

Magnet position behavior

- Energizing coil A opens the magnet; the Hall voltage falls to roughly 0.5–0.6 V (OPEN condition < 700 mV).
- When coil A is de-energized, the valve spool returns to center but the magnet stays open.
- Energizing coil B reverses the magnet to close it; the Hall voltage rises toward 2.5 V (CLOSE condition > 2300 mV).

WIRING & I/O — M12 CONNECTORS

Connector A — Outputs to valve coils (M12 5-pin, common-positive / sinking)

Pin	Wire	PLC terminal	Component	Function
Pin 1	+24 V	24V	—	Supply common (+24 V); 24 V is split to the + terminals of coils A / B / C.
Pin 2	0V A	Y1	SOL_A	Directional valve A coil (magnet OPEN).
Pin 3	0V B	Y2	SOL_B	Directional valve B coil (magnet CLOSE).
Pin 4	0V C	Y3	SOL_C	Relief valve C coil (MTC only).
Pin 5	—	—	—	Spare (not wired).

Connector B — Inputs from two 270758 Hall sensors (M12 5-pin)

Pin	Wire	PLC terminal	Component	Function
Pin 1	+24 V	24V	—	Shared supply for both Hall sensors.
Pin 2	0 V	GND	—	Power and analog ground (must be common with analog GND).
Pin 3	Signal 1	AD0	D8030	Hall sensor 1 analog signal (0–5 V).
Pin 4	Signal 2	AD1	D8031	Hall sensor 2 analog signal (0–5 V).
Pin 5	—	—	—	Spare (not wired).

Notes

- Each output switches the coil low (to 0 V); the COM side connects to 0 V (sinking outputs).
- Analog scale: 0–10 V at the input maps to 0–4096 counts; displayed mV = AD × 10000 / 4096.
- No discrete inputs and no analog outputs are used. The Hall sensors are 24 V, 0–5 V, 3-wire devices.

ALARMS & TROUBLESHOOTING

Alarm conditions

Condition	Indication	Action / reset
E-STOP pressed	All outputs (Y1/Y2/Y3) off; ALARM lamp flashes.	Remove the cause, then press ALARM RESET.
Position timeout (closed-loop)	Sensor does not reach the threshold within the hold time; cycle stops; ALARM lamp flashes.	Check valve/sensor and hold times, then ALARM RESET and restart.
Sensor disconnected	Reading near 0 mV (< 100 mV); OPEN will not falsely trigger.	Reconnect the sensor, then ALARM RESET.
Target count reached	Cycle stops at TARGET COUNT (normal).	Press RESET COUNT to start a new run.
Reset counter	—	RESET COUNT stops the cycle and clears the counter.
Alarm lamp pattern	1 s on / 1 s off for sensor faults and E-STOP.	—

Troubleshooting

Symptom	Possible cause	Remedy
MAGNET OPEN does not light	Sensor below open threshold not reached; sensor disconnected.	Check coil A and wiring; verify sensor voltage and the 700 mV threshold.
MAGNET CLOSED does not light	Coil B not reversing; close threshold not reached.	Check coil B and wiring; verify the 2300 mV threshold.
Cycle stops immediately	Closed-loop timeout; no position signal.	Test in OPEN loop to confirm motion, then check sensors and hold times.
ALARM lamp flashing	Latched sensor fault or E-STOP.	Clear the fault, then press ALARM RESET.
VALVE C / JOG C missing	MAC mode selected.	Switch to MTC mode.

WARNINGS



Review and understand all warnings before operating the unit.
Contact and consult Magswitch engineering at +86 182 6817 6813 if you have any questions or concerns about your application or the tool's operation.

- Operate the unit only with a 24 V DC supply. All wiring and connector pin-out must be performed by qualified personnel.
- Do not connect or disconnect the M12 connectors while the unit is powered.
- The HMI E-STOP is an emergency output cut-off; it is not a substitute for machine guarding or a safety-rated protection system.
- De-energizing coil A does not close the magnet — the spool returns to center and the magnet remains open until coil B is energized. Plan material handling and guarding accordingly.
- In closed-loop mode, do not run the cycle until the Hall sensors are confirmed and the OPEN / CLOSE thresholds are set correctly.
- Do not operate the unit beyond its rated temperature or enclosure rating.
- Always test the complete installation before deployment.

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